

Farzad Avishan

F.avishan@gmail.com | +1 514 233 4685 | LinkedIn | Google Scholar

PROFESSIONAL SUMMARY

Operations research and data scientist specializing in optimization, predictive modeling, and data-driven decision support for complex operational problems. Currently a Research Scientist in Data & AI at Air Canada, with prior postdoctoral research experience at HEC Montréal and a Ph.D. in Industrial Engineering. Experienced in developing optimization models, analytical frameworks, and scalable solution approaches for airline operations, logistics, supply chain, electric mobility, production planning, and transportation applications.

PROFESSIONAL EXPERIENCE

Research Scientist, Data & AI Air Canada, Canada Oct 2024 – Present

- Develop decision-support methodologies that translate operational requirements into implementable optimization and analytical solutions.
- Apply mathematical optimization and analytical techniques to support complex business and operational decision-making.
- Develop repeatable processes for large-scale data analysis, optimization-model development, validation, and implementation.
- Leverage LLMs and agentic AI to develop intelligent workflows, coding agents, and automation solutions that improve analytical and software-development productivity.

Postdoctoral Research Fellow HEC Montréal, Canada Oct 2023 – Oct 2024

- Developed mathematical optimization models and solution approaches for transportation, logistics, routing, and electric-mobility applications.
- Conducted computational research on large-scale combinatorial optimization problems and evaluated algorithmic performance.
- Collaborated with researchers and industry stakeholders to translate operational problems into quantitative decision-support models.

Research Assistant Ozyegin University, Turkey Sep 2022 – Oct 2023

- Compiled and analyzed research results and contributed to scholarly publications and research documentation.
- Designed optimization models and solution approaches for production planning, logistics, and supply chain applications. transportation, logistics, routing, and electric-mobility applications.
- Performed literature reviews, computational research, and analytical experiments to support academic studies.

SELECTED APPLIED PROJECTS

Transportation Cost Forecasting Clear D3 March–September 2023

- PI: İhsan Yanıkoğlu; co-PIs: **Farzad Avishan** and Berk Karasu. “Utilizing Historical Data to Develop a Predictive Model for Forecasting Transportation Costs in Upcoming Periods.” Funded by Clear D3, March–September 2023.

TUBITAK projects Turkey, Istanbul 2018 – 2024

- PI: İhsan Yanıkoğlu; co-PIs: Basak Altan and Mehmet Soysal. “Applications of Optimization under Uncertainty Methods on Production Planning, Supply Chain and Logistics Problems.” Funded by TUBITAK-3501, February 2022–February 2024.
- PI: Okan Örsan Özener; co-PIs: Basak Altan and Ali Ekici. “Cooperative Game Theory Approaches for Collaborative Dynamic Lot Sizing Problem.” Funded by TUBITAK-1001, November 2018–November 2021.

EDUCATION

Ph.D., Industrial Engineering Özyeğin University, Istanbul, Turkey Mar 2023

- Research focus: robust optimization and optimization under uncertainty for logistics, transportation, supply-chain, and production-planning applications.
- Thesis: *Applications of Robust Optimization in Logistics and Production Planning*. CGPA: 3.87/4.00.

M.Sc., Industrial Engineering Mazandaran University of Science and Technology, Iran Sep 2016

- Thesis: *A New Optimization Model for Production and Distribution of Supply Chain with Green Logistic Approach*.

B.Sc., Industrial Engineering Tabriz University, Iran Sep 2014

CORE STRENGTHS & TECHNICAL SKILLS

Optimization & analytics: mathematical optimization, combinatorial optimization, robust optimization, optimization under uncertainty, heuristic algorithms, predictive modeling, forecasting, exploratory data analysis, model validation, decision-support systems, large-scale network optimization.

Programming & software: Python, C++, Gurobi, FICO Xpress, CPLEX, GAMS, MATLAB, Git, GitHub.

Applied domains: airline operations, transportation, logistics, supply chain, electric vehicles, production planning, humanitarian logistics, and network optimization.

AI development: LangChain, LangGraph, AI agents, and AI-assisted software-development workflows.

PUBLICATIONS

Selected publications and citation information: [Google Scholar Profile](#)